SOCKET PROGRAMMING

Socket Programming:

Socket programming is a way of connecting two computers on a network to communicate with each other.

Network:

A set of co operative interconnected computers is called as network.

Sockets make the transfer of information possible between two different programs or devices.

Client Socket:

A client socket is a connection end at client side.

Server Socket:

A server socket is a connection end at server side.

Port Number:

It is used to identify the service over network or internet. Port numbers range is 0 to 65535.

Reserved port numbers range is 0 to 1023 whereas free port numbers range is 1024 to 65535.

To achieve socket programming in python, we need the following methods of socket module.

1) socket() => Used to create sockets at both client side and server side.
2) accept() => Used to accept the connection. It returns socket object and address.
3) bind() => Used to bind the address and port number.
4) close() => Used to close the socket.
5) listen() => Used to enable the server to listen connections.
6) gethostname() => Used to get the host name.

7) send() => Used to send the information.
8) recv() => Used to receive the information.
9) connect() => Used to establish a connection.

Examples:

1) Program to get the ip address of the system:

```
import socket
ip=socket.gethostbyname("www.google.com")
print(ip)
```

2) Program to send the request to convert the given string into capital letters to server:

Server Program:

```
from socket import *
s=socket()
s.bind(("localhost",2000))
s.listen(1)
c,add=s.accept()
msg1=c.recv(1000)
msg2=msg1.decode("utf-8")
msg3=msg2.upper()
msg4=msg3.encode("utf-8")
c.send(msg4)
s.close()
```

Client Program:

```
from socket import *
s=socket()
s.connect(("localhost", 2000))
msg1=input("Enter any message in lower case letters: ")
msg2=msg1.encode("utf-8")
```

```
s.send(msg2)
msg3=s.recv(1000)
msg4=msg3.decode("utf-8")
print("From Server: ", msg4)
s.close()
```

3) Program to demonstrate chatting program:

Server Program:

```
from socket import *
s=socket()
s.bind(("localhost",2000))
s.listen(1)
c,add=s.accept()
while True:
        msg1=c.recv(1000)
        msg2=msg1.decode("utf-8")
        print("From Client: ", msg2)
        msg3=input("To Client: ")
        msg4=msg3.encode("utf-8")
        c.send(msg4)
```

Client Program:

```
from socket import *
s=socket()
s.connect(("localhost", 2000))
while True:
        msg1=input("To Server: ")
        msg2=msg1.encode("utf-8")
        s.send(msg2)
        msg3=s.recv(1000)
        msg4=msg3.decode("utf-8")
        print("From Server: ", msg4)
```